Case Report/Case Series

Management of Occult Necrotic Melanomas

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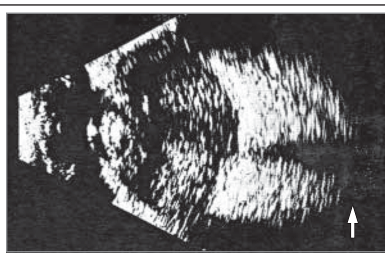
As many as 20% of uveal melanoma–containing eyes historically had opaque media and more than half of such tumors were unsuspected.[1,2] While many ophthalmic oncologic centers, including ours, have more than 99% accuracy of uveal melanoma diagnosis, in eyes with opaque media and extensive necrosis the correct diagnosis can sometimes be impossible to establish with ultrasonography, magnetic resonance imaging (MRI), and fine-needle aspiration biopsy (FNAB).[3,4] We present 5 cases that illustrate these problems.

Report of Cases

Case 1

A 72-year-old man had a left retinal inferior detachment and a dark mass; prior ultrasonography was thought to show a reabsorbing hemorrhage. He presented in 1988 with 10 days of uniocular pain. On examination, his visual acuity was light perception with a pressure of 38 mm Hg. He had corneal blood staining. We could not visualize the ocular interior. Ultrasonography could not differentiate a hemorrhage from a neoplasm (**Figure 1**). The eye was enucleated and showed ischemic necrosis of the iris, ciliary body, and retina, calcification and bone formation, intraocular hemorrhage, and a tumor with more than 90% necrosis. At 5 years, no metastases developed.

Figure 1. Case 1



Ultrasonography demonstrating an indeterminate lesion. We could not differentiate between a hemorrhagic process and a neoplasm.

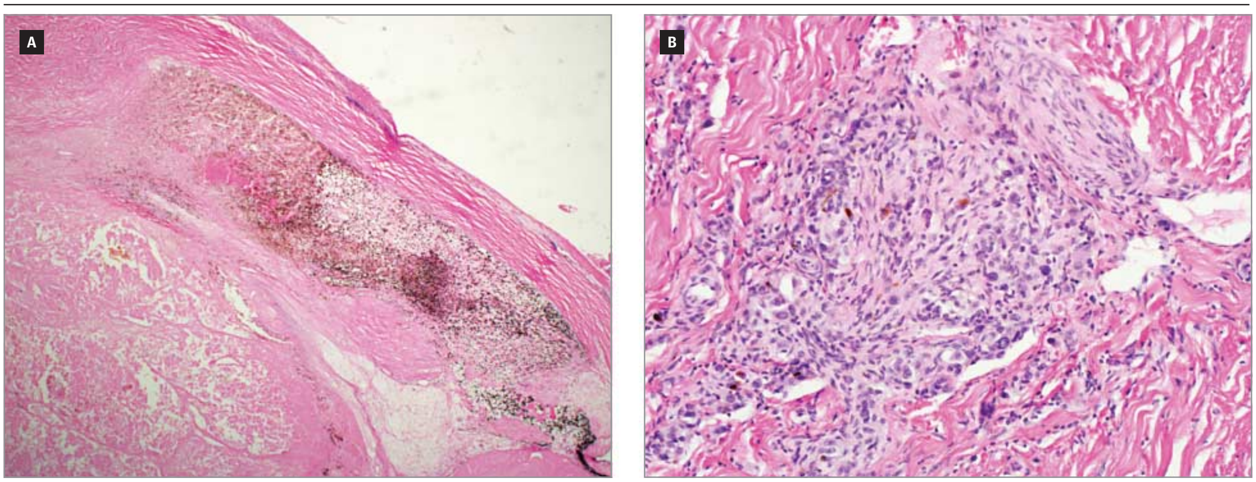
Case 2

A 69-year-old man had decreased vision for 4 years and initially refused evaluation and treatment. Three weeks prior to referral he noted a red eye. On our examination in 1986, his visual acuity was no light perception in the involved eye with 3+ chemosis. There was flocculant material in the anterior chamber and iris neovascularization. We could not visualize the involved fundus. Tumor could not be diagnosed on ultrasonography; A-scan showed an area of medium to high internal heterogeneity without spontaneous pulsation or attenuation. The enucleated eye showed neovascular angle closure and a tumor from the ciliary body to the optic nerve, almost totally necrotic, with just 2 small areas of identifiable melanoma cells with extrascleral extension (**Figure 2**). Metastatic disease was diagnosed 1 year later.

Case 3

A 70-year-old woman developed left eye pain. Vitreous hemorrhage and increased pressure led to a vitrectomy and a possible mass was noted. She was referred in 1995 with light perception visual acuity, cataract, posterior synechiae, and vitreous hemorrhage. We could not visualize a mass. An MRI

Figure 2. Case 2



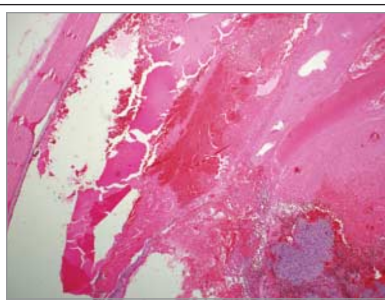
A, Necrotic melanoma in the anterior segment (hematoxylin-eosin, original magnification ×2). B, A few identifiable melanoma cells in a posterior scleral canal (hematoxylin-eosin, original magnification ×20).

and ultrasonography were not diagnostic and FNAB results under ultrasonic guidance were normal (only red cells and some inflammatory cells were seen). The enucleated eye showed neovascular angle closure and a mixed spindle and necrotic melanoma extending into the vitreous with more than 90% necrosis (**Figure 3**). She was diagnosed with metastatic disease 3 years later.

Case 4

A 58-year-old Chinese woman had a blind, painful eye. A prior B-scan showed an organized vitreous hemorrhage; a vitreous tap did not show malignancy. She was given a retrobulbar alcohol injection for eye pain. On our examination in 1988, she had corneal blood staining and a pressure of 50 mm Hg. Ultrasonography was not diagnostic and FNAB results were normal (only inflammatory cells and red cells were noted). The enucleated eye showed neovascular angle closure, ischemic necrosis of the iris and ciliary body, and a diffuse, necrotic pigmented tumor with only a few identifiable melanoma cells adjacent to and extending into the optic nerve. She died of unknown causes 10 years later.

Figure 3. Case 3



Necrotic melanoma with a small area of identifiable melanoma cells in the lower right corner of the photomicrograph (hematoxylin-eosin, original magnification ×2).

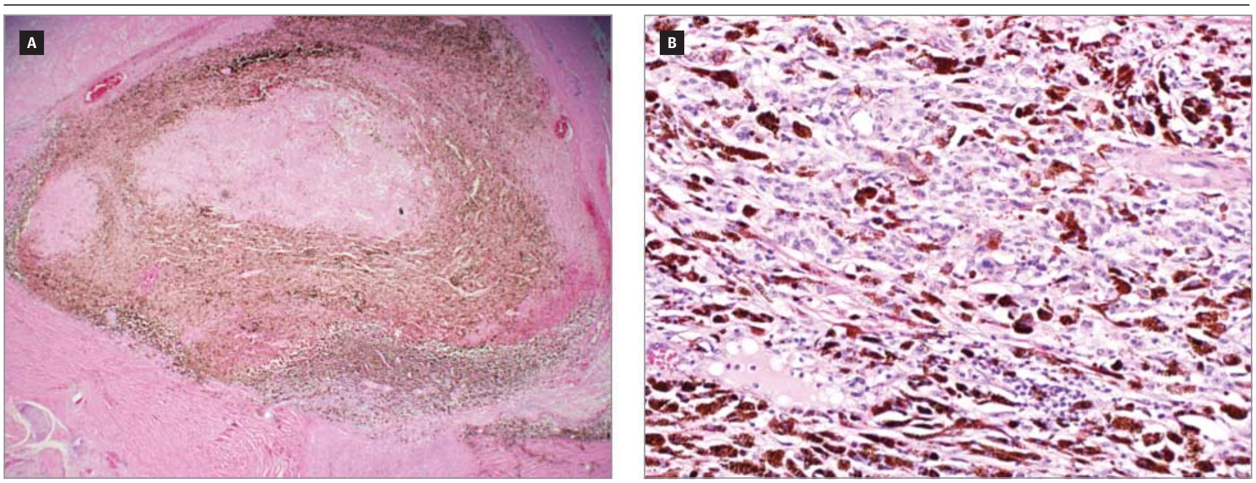
Case 5

A 69-year-old man was referred in 2012 with a chronically blind, painful left eye and hyphema. Ultrasonography and computed tomography did not show obvious tumor. On examination, his visual acuity was no light perception and we could not visualize the fundus. Results of FNAB done in the center of the area identified on ultrasonography were normal and did not show tumor cells (only red cells and a few inflammatory cells were noted). The enucleated eye showed neovascular angle closure, ischemic iris necrosis, a total retinal detachment, and a mostly necrotic melanoma over the optic nerve and into the sclera around a short posterior ciliary vessel (**Figure 4**). He is presently alive with no evidence of metastatic disease.

Discussion

Some ophthalmologists use evisceration in the management of blind eyes with opaque media. Concern has been raised about cases in which an undiagnosed uveal melanoma was eviscerated. In an article by Eagle et al,[5] several patients had eviscerations with an undetected uveal melanoma and minimal preoperative ophthalmic oncologic evaluations. An inference of that article was that had the patients been evaluated with ophthalmic oncologic tests, some of them would have been correctly diagnosed and an enucleation would have been performed. Unfortunately, in our experience, there is a very small subset of patients (<0.1% of our melanoma cases) with a blind, painful eye with opaque media and necrotic tumors in whom even a complete ophthalmic oncologic evaluation, including imaging with ultrasonography and MRI as well as intraoperative FNAB, does not establish the correct diagnosis. We therefore take their argument a step further; in cases of a blind, painful eye in which the posterior pole cannot be evaluated and a tumor ruled out, the eyes should be enucleated and not eviscerated.

Figure 4. Case 5



A, Necrotic melanoma in the posterior pole over the optic nerve with a few identifiable spindle B cells at the base (hematoxylin-eosin, original magnification ×2). B, Higher-power view of spindle B melanoma cells (hematoxylin-eosin, original magnification ×20).

The common pathologic features of all these cases were a necrotic melanoma with only a few histologically identifiable melanoma cells. The area of identifiable, nonnecrotic melanoma cells was too small to produce a characteristic melanoma pattern on ultrasonography or MRI and too small to localize for a site to perform an accurate FNAB. The chances of an FNAB missing the few identifiable cells were overwhelming.

Furthermore, if an evisceration had been performed, the chances of finding these cells on the submitted fragments of the tissue would be low since serial sections are not usually performed on this tissue. In 1 case, the identifiable cells were in a sclera canal, which would not have been removed in an evisceration. Therefore, the diagnosis of a melanoma could easily be missed, to the detriment of the patient.

While ophthalmic oncologic diagnosis is extremely accurate in most uveal melanomas, in cases in which there is an eye with no visualization of the tumor, with significant hemorrhage and necrosis present, we cannot reliably diagnose a uveal melanoma by any available technique except histological examination. For these reasons, we continue to emphasize that enucleation of such eyes is the safer option.

ARTICLE INFORMATION

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